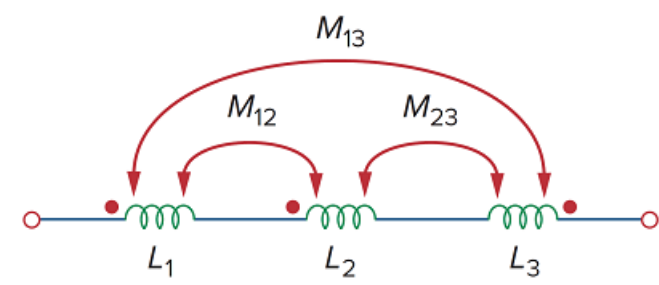


Problem

Using Fig. 13.73, design a problem to help other students better understand mutual inductance.

Figure 13.73



Step-by-step solution

Step 1 of 2

Refer to Figure 13.73 in the textbook.

The three inductors are connected in series. So, current through all inductor is same.

First consider the inductor, L_1 . The mutual inductor with inductor, L_2 is positive as both the current is entering in both inductors L_1 and L_2 . The mutual inductor with inductor, L_3 is negative as the current is entering in inductor, L_1 and the current is leaving in inductor, L_3 .

The equivalent inductor across inductor, L_1 is,

$$L_{1eq} = L_1 + M_{12} - M_{13}$$

Step 2 of 2

Now consider the inductor, L_2 . The mutual inductor with inductor, L_1 is positive as both the current is entering in both inductors L_1 and L_2 . The mutual inductor with inductor, L_3 is negative as the current is entering in inductor, L_2 and the current is leaving in inductor, L_3 .

The equivalent inductor across inductor, L_2 is,

$$L_{2eq} = L_2 + M_{12} - M_{23}$$